



Tribhuvan University Faculty of Humanities and Social Science

Real Estate Management System

A PROJECT REPORT

Submitted to

Department of Computer Application

Birendra Multiple Campus

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by

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Supervisor's Recommendation

I hereby recommend that this project prepared under my supervision by **MEWALAL LONIYA** entitled “**REAL ESTATE MANAGEMENT SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Application is recommended for the final evaluation.

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Tribhuvan University
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LETTER OF APPROVAL

This is to certify that this project prepared by **MEWALAL LONIYA** entitled “**REAL ESTATE MANAGEMENT SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor in Computer Application has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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ABSTRACT

The project aims to design & implementation of an real estate management system which will provide customers a facility to buying and selling property in online without any hassle. It is a web based system & it will save valuable time and effort of customer. The real estate management system designed in this project was developed using PHP, JavaScript, CSS and html as the programming languages and MySQL as the database Management system. The methodology adopted for this project is the waterfall methods.

This system allows the customer to cancel his/her reservation, if any problem occurs and view price chat which did not occur in the existing system. This system allows the customers to search for property that are available between the many cities, for a particular property and dream project. The system displays all the properties details such as property location, name, property types, status, price etc. After search the system display list of available property and allows customer to choose a particular property. Then the system checks for the availability of another types of related properties. Then it checks the property and buy sell.

Acknowledgement

We are extremely grateful and remain indebted to almighty god who has guided in all ventures to complete our project successfully.

We would like to thank our family who always support us in our study. They were the reason that keeps us go further and to be successful in our life.

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1 Introduction

1.1 Introduction

Real Estate Management System (rems) is web-based application. that takes care of everything that a real estate company wants to do. Property Management System is developed for real estate Companies. It is very strong and easy to use that makes quick properties and account handling process. The real estate Property management system (rems) is web-based software and you can access it from anywhere.

The application is easy to use and has user-friendly features and functionalities. It also has a simple and pleasant user interface to give end-users a better experience while using the web application.

1.2 Problem Statement

- Customer should go out for searching properties.
- It is time taken to visiting many place for property deal.
- Many difficulties are faced during the searching a good property.

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1.3 Objective

The objective of this project is to study how the Real estate management system website works and the process of the system followed by the users.

- Real estate management system (rems) can provide many dream properties without going out of the home.
- (rems) can be save your money and time for searching dream properties.
- Shows your ability to market yourself as a real estate agent.
- Allows you to highlight your best professional attribute.

1.4 Scope and Limitation

1.4.1 Scope

We believe that Real estate management system will become even more popular in the future for the obvious reason that they save users, time and money by visiting the site and take a look for which property is available and also agent will be informed and they can easily discuss about property. Urban areas are well equipped to provide the facilities that we want to provide. Our project suits in the area where there is availability of many houses building and no issues of internet access.

1.4.2 Limitation

- Internet accessibility
- Low accuracy
- Small system so it can't store more data.

1.5 Report Organization

This system will be implemented with the responsive UI design for better user experience. This system is very simple & user friendly so, any user can use this system easily. At present there are many Real estate management system is available in Nepal.

Reading the report from introduction to conclusion is quite monotonous task so, I came with an idea to write short and fast way to describe from chapter 1 – 5 continuously.

Firstly, I give an introduction about what is Real estate management system (rems)and how it helps the customer to find properties and used to buy property as well as they are required through online. How this software works and describes about the main objective of this system to solve the problems.

Secondly, In background study and literature review, I wrote how I came into conclusion of developing an application by doing the research in different online platform related to

sopping market. How this can help the owner to operate their business efficiently without any errors to get the product even without going out from room.

Thirdly, In system analysis and design what kind of approach and design I have used is shown via screenshot and table. Moreover, an admin can add, manage and delete property category, properties details and owner details. There is also the structures of database design which will help the admin to view the listed category, agent, user and properties.

Furthermore, In case of implementing and testing while designing the ER, DFD and other figure I used draw.io and visual paradigm tool which was an online tool that will provide a free drawing using browser with easy interface. There was two phase- Unit and System testing where unit testing was conducted for admin module, user module. If admin, user tend to change username and password they can change it.

Finally, I concluded with how this software changes the traditional method of buying property into digital method. How this become convenient and what kind of feature this software lack which can be added later so, it will perform better than the previous version.

2 Background Study and Literature Review

2.1 Background Study

We implement this system for better user experience, because real estate is a big hassle for users now. This system is very easy to access. Also for establishing real time communication, using modern and updated technology. So, users can see the update without reload or refresh. This system will be compatible with user devices such as PC, laptop, tablet & smart phone. So users can easily access the system anytime anywhere. This system will be implemented with the responsive UI design for better user experience. This system is very simple & user friendly so, any user can use this system easily. At present there are many real estate management systems available in Nepal.

2.2 Literature Review

Real estate management System is the system where we can buy the property through internet with just one click, which can make our life easy and amazing. Presently the customer spends a lot of days going to the searching dream property.

Some of the customers don't always get the selection they want because the agents run out of certain property or because there is no provision of buying custom property.

Through our research we determined that the system that exists in today's date is not so efficient like if we want to buy dream property then we need to call to agents than ask for the good property and soon, that is too long process and not trust worthy too and there may be lots of confusion which may not build a good relation between agents and the property. We are not saying that this process is totally wrong or useless but we can make it better and trust worthy through internet (Real estate management) system. Where we can see our dream property with its actual price, which will avoid a long quotes and confusions of both agents and the customers too.

3 System Analysis and Design

3.1 System Analysis

I used the Waterfall methodology while building this application. This project had fixed specification, ample time and enough resources so Waterfall methodology was used to build this system. (1)

Software development life cycle process specifies a method of developing the software. Each software development projects start with some needs and ends with some software that satisfies those needs. A software development life cycle specifies the set of activities that should be performed to go from user needs to final products. I have been working with a fixed requirement so I had created fixed milestones and deadlines. Waterfall methodology fitted perfectly with those requirements

3.1.1 Requirement Analysis

It is the process of defining the expectations of the users for an application that is to be built or modified. It needed to fulfill functional and non-functional requirement. (2)

3.1.1.1 Functional Requirement

Functional requirements define the capabilities and functions that a system must be able to perform successfully. The functional requirements of this Real estate management system include:

- Admin can login and logout.
- Admin can add, update and delete properties.
- Admin can add, update and delete properties details and other details.
- Admin can list, view, update and delete features of properties.
- User can login, logout and manage own detail.
- Costumer can login and logout.
- Customer can and manage own detail.
- Customer can list available properties, like, dislike, and give feedback

3.1.1.2 Non-functional Requirement

A non-functional requirement is a requirement that specifies criteria that can be used to judge the operation of a system, rather than specific behaviors.

Some of the non-functional requirements include:

Backup- provision for data backup

Maintainability- easy to maintain

Performance/ response time- fast response

Usability by target user community- easy to use

Expandability- needs to be future proof or upgradable.

3.1.2 Feasibility Analysis

A feasibility study is a test of a system proposal. According to its work ability, impacts on the organization, ability to meet user needs and effective use of the resources its main task done during feasibility study are

3.1.2.1 Technical Feasibility

Technical feasibility this project will be focused on gaining an understanding of the present technical resources of the organization and their applicability to the expected needs of the proposed system. It is an evaluation of the hardware and software and how it meets the needs of the proposed system.

Technology Used:

Operating system: Windows XP / windows 7

Server: Apache 2.4.4

Front-end: PHP 5.4.19 (Framework: Bootstrap)

Back-end: MySQL 5.5.32(using phpMyAdmin 4.0.6)

We have used XAMPP which is a free and Open Source Cross-Platform Web Server Solution stack. It comes with Apache Web Server, MySQL Database and PHP.

3.1.2.2 Operational Feasibility

System is operational feasible since the user are familiar with the technologies and hence there is no need to gear up the personnel to use the system.. In Operational Feasibility service requirements study is done, along with how much easy product will be to operate and maintenance after deployment. Along with this other operational scopes are determined usability of service, determining suggested solution by software development team is acceptable. The system is operational feasible since the user are familiar with the technologies and hence there is no need to gear up the personnel to use the system. Also the system is very user friendly and easy to use.

3.1.2.3 Economic Feasibility

In this Feasibility study cost and benefit of the project is analyzed. Means under this feasibility study a detail analysis is done, cost of the project for development which includes all required cost for final development like hardware and software resource required, design and development cost and operational cost and so on. After that it is analyzed, project will be beneficial in terms of finance for organization.

3.1.2.4 Expected Working Schedule

Proposed system would be designed as per the time calculation. Likelihood that time frames can meet their adequate to meet organization's needs.

TIME PHESE	JAN	FEB	MARCH	APIR	MAY	JUNE
STUDY AND ANALYSIS	1 WEEK	2 WEEK				
DATA FACHTING AND SCRAPPING	1 WEEK					
COADING AND IMPLEMENTION			2 WEEK	3 WEEK		
TESTING						
DOCUMENTAION					2 WEEK	
REVIEWS						1 WEEK
PRESANTAION	1 DAY					

Figure 3-1 Gantt Chart

3.1.5 Process Modeling (DFD)

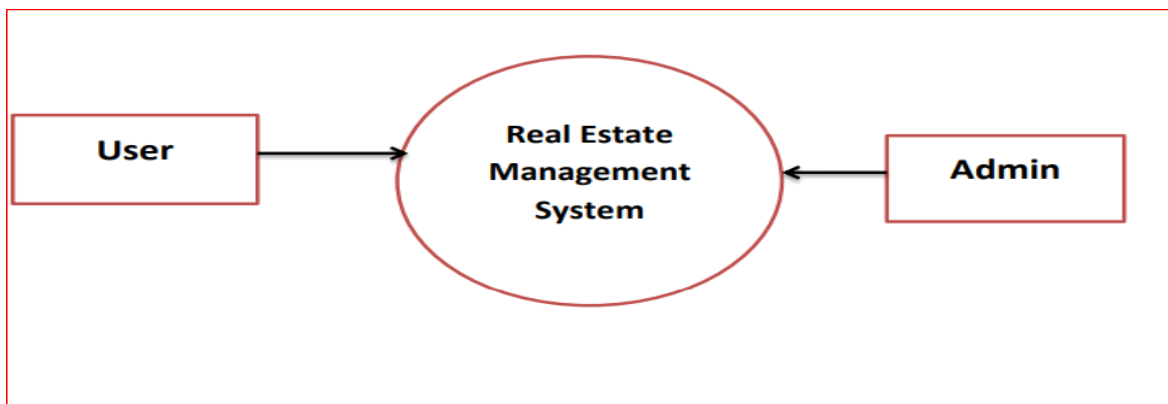


Figure 3-2 DFD Level – 0

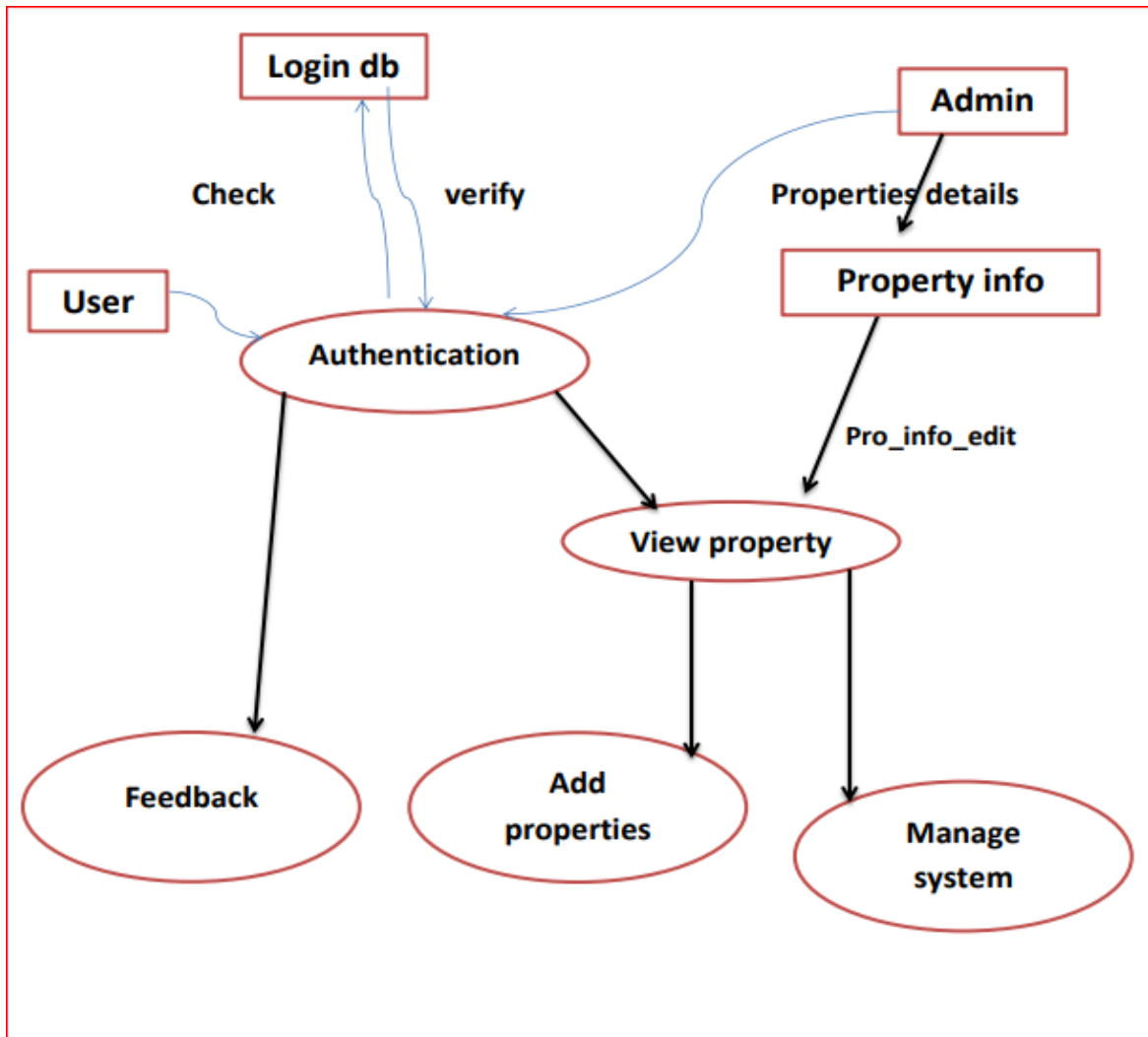


Figure 3-3 DFD Level – 1

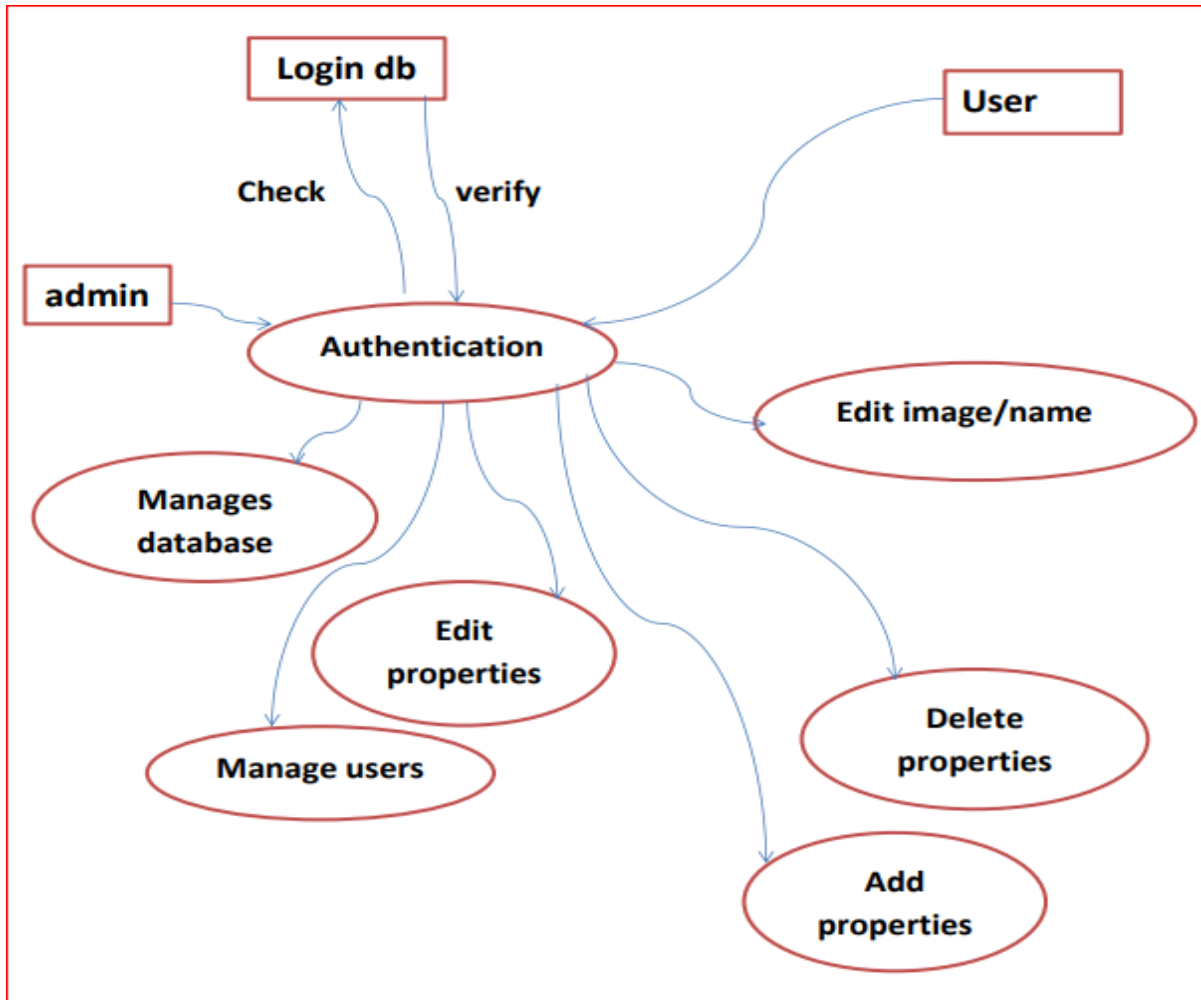


Figure 3-4 DFD Level – 2

3.2 Data Modeling (ER-Diagram)

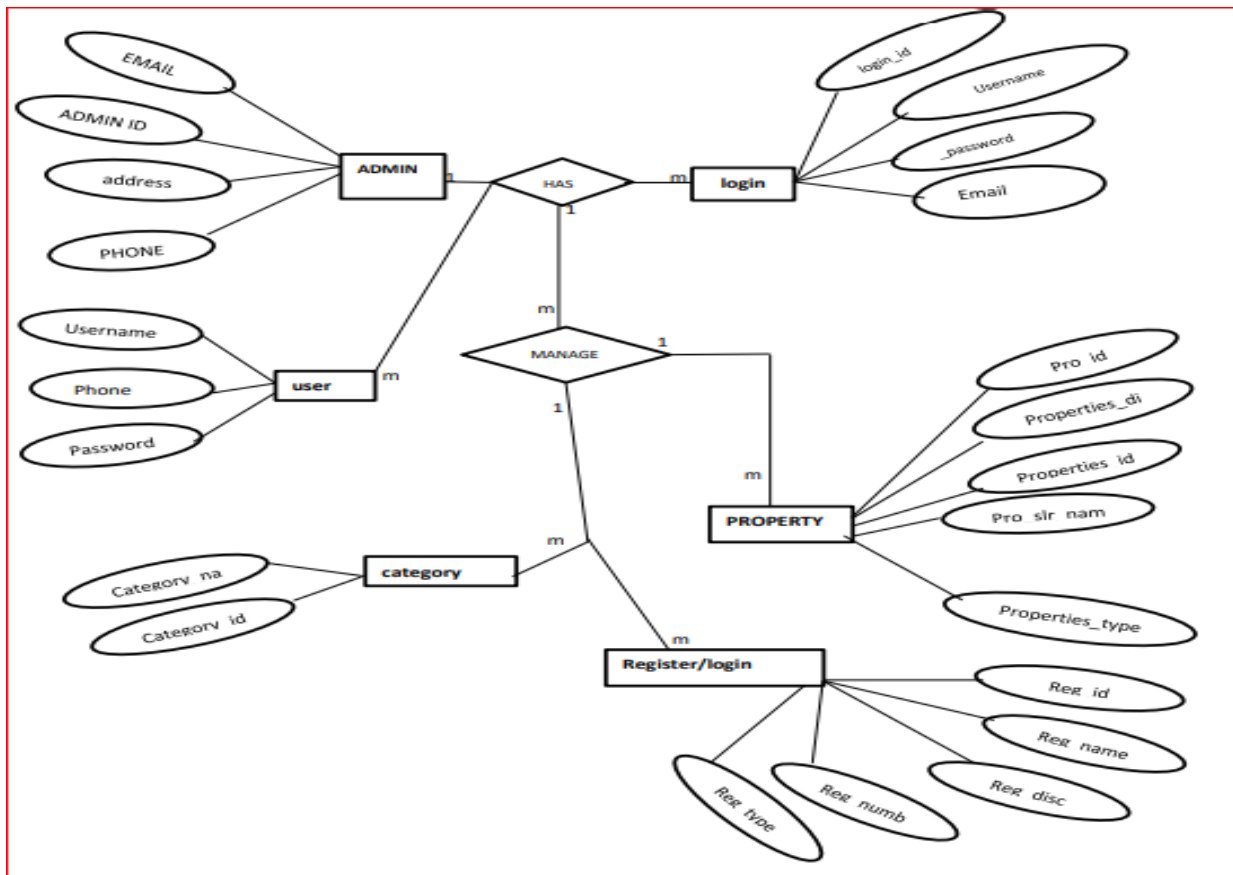


Figure 3-5 ER diagram

3.2.1 Architectural Design

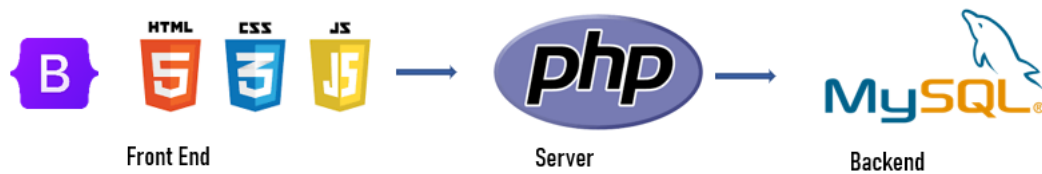


Figure 3-6 System Architecture

3.2.2 Database Schema Design

The general theme behind a database is to handle information as an integrated whole. A database is a collection of inter-related data stored with minimum redundancy to serve single users quickly and efficiently. The general objective is to make information necessary, quick, inexpensive and flexible for the user

3.2.2.1 Database Table

Field Name	Data Type	Size	NULL	Constrain
Id				
Title	int	30	No	Primary Key
Area	varchar	100	Yes	
Address	text	-	Yes	
Discription	Varchar	255	Yes	
image	Varchar	255	Yes	
Sold/rent	Varchar	255	Yes	
kechen	Varchar	255	Yes	
Hall	Varchar	255	Yes	
Bedroom	Varchar	255	Yes	
bathroom	Varchar	255	Yes	
balcony	Varchar	255	Yes	

Table 3-7 Properties Table

Field Name	Data Type	Size	NULL	Constrain
Id	int	30	No	Primary Key
Username	varchar	250	Yes	
email	varchar	250	Yes	
Username	text	-	Yes	
Password	text	-	Yes	

Table 3-8 Register Table

Field Name	Data Type	Size	NULL	Constrain
Id	int	30	No	Primary Key
Username	varchar	250	Yes	
email	varchar	250	Yes	

Table 3-9 Admin Table

Field Name	Data Type	Size	NULL	Constrain
Id	int	30	No	Primary Key
Username	varchar	250	Yes	
email	varchar	250	Yes	

Table 3-10 Properties Type Table

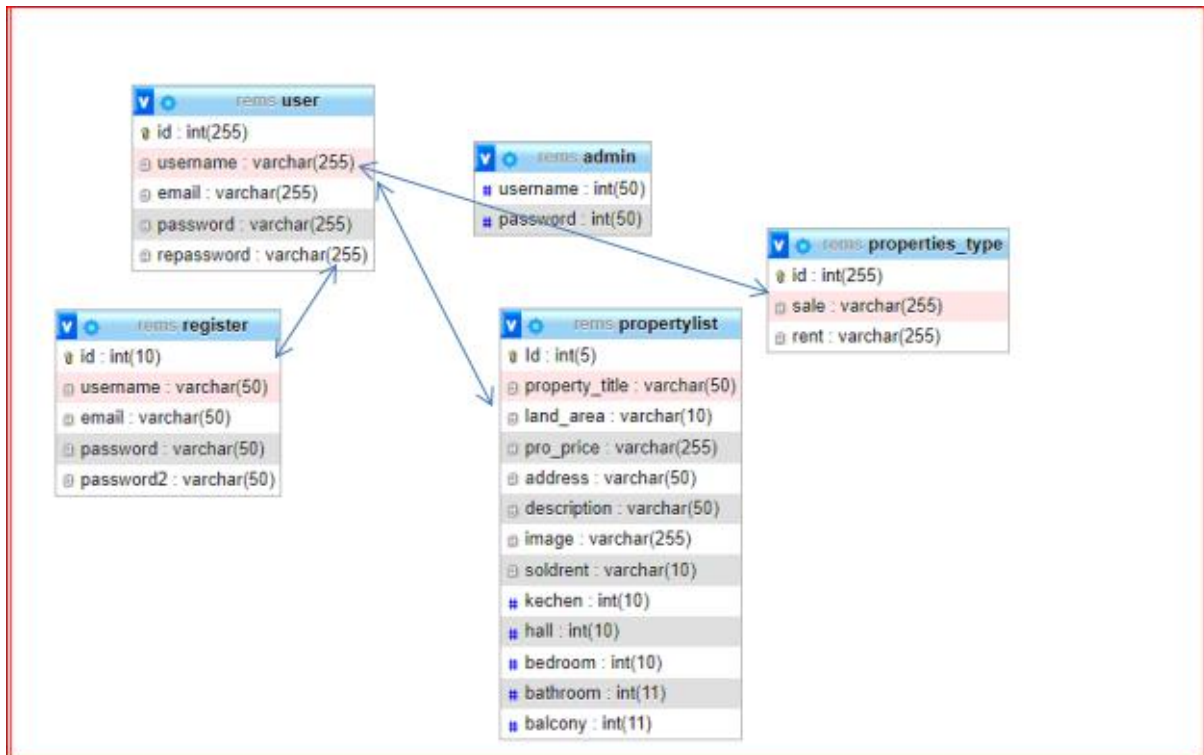


Table 3-11 Database Table

3.2.3 Interface Design (UI Interface / Interface Structure Diagrams)

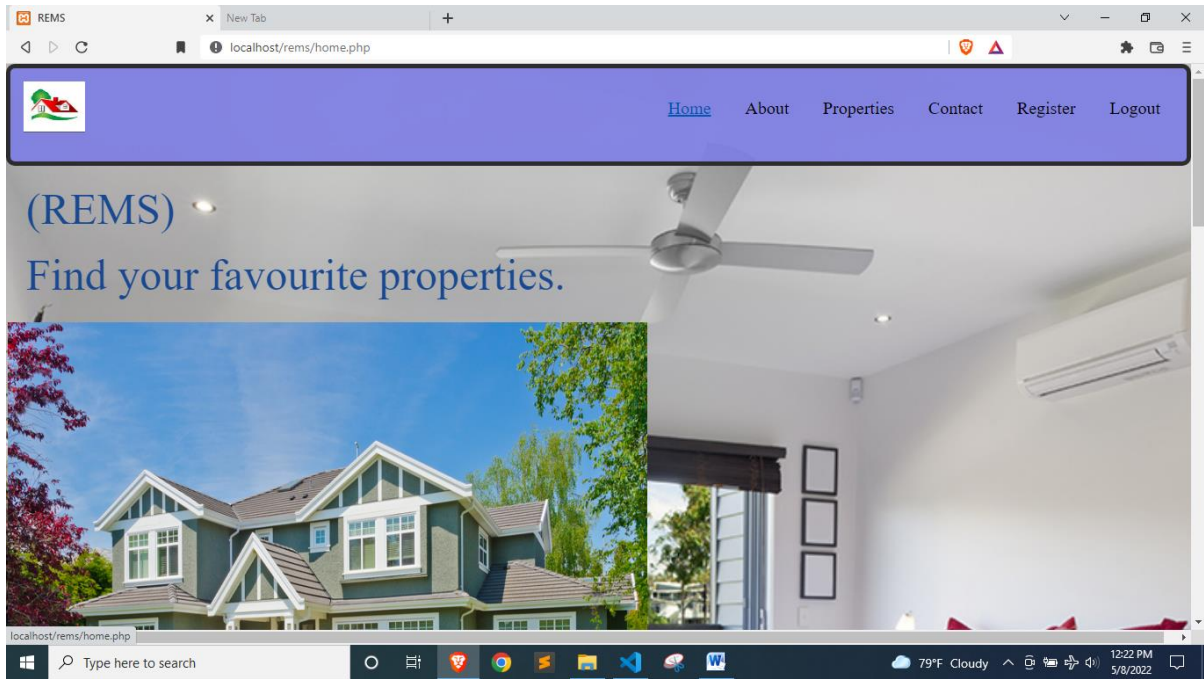


Figure 3-12 Home page Interface

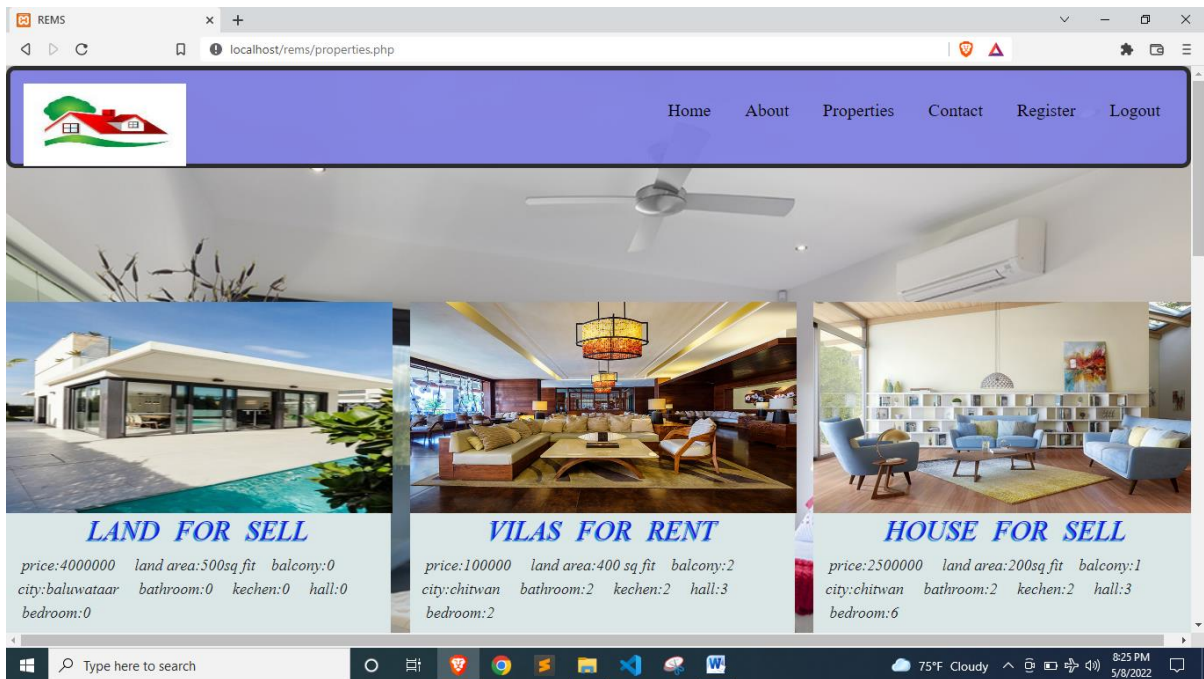


Figure 3-13 Available Properties Interface

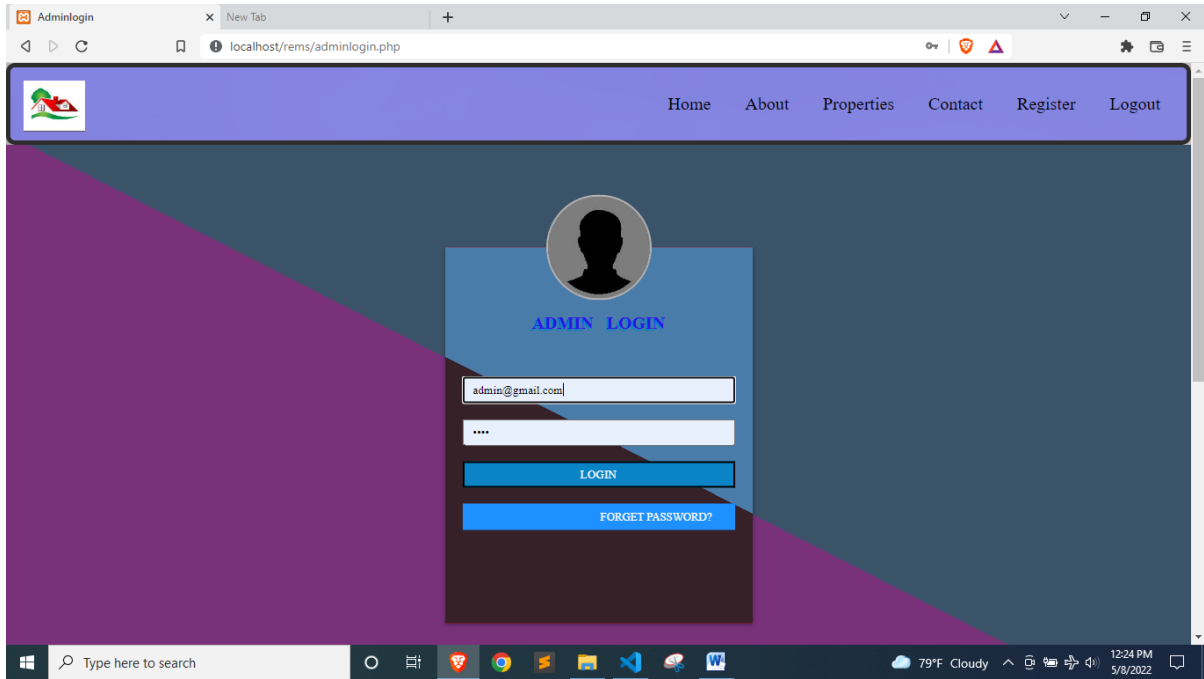


Figure 3-14 Admin Login Interface

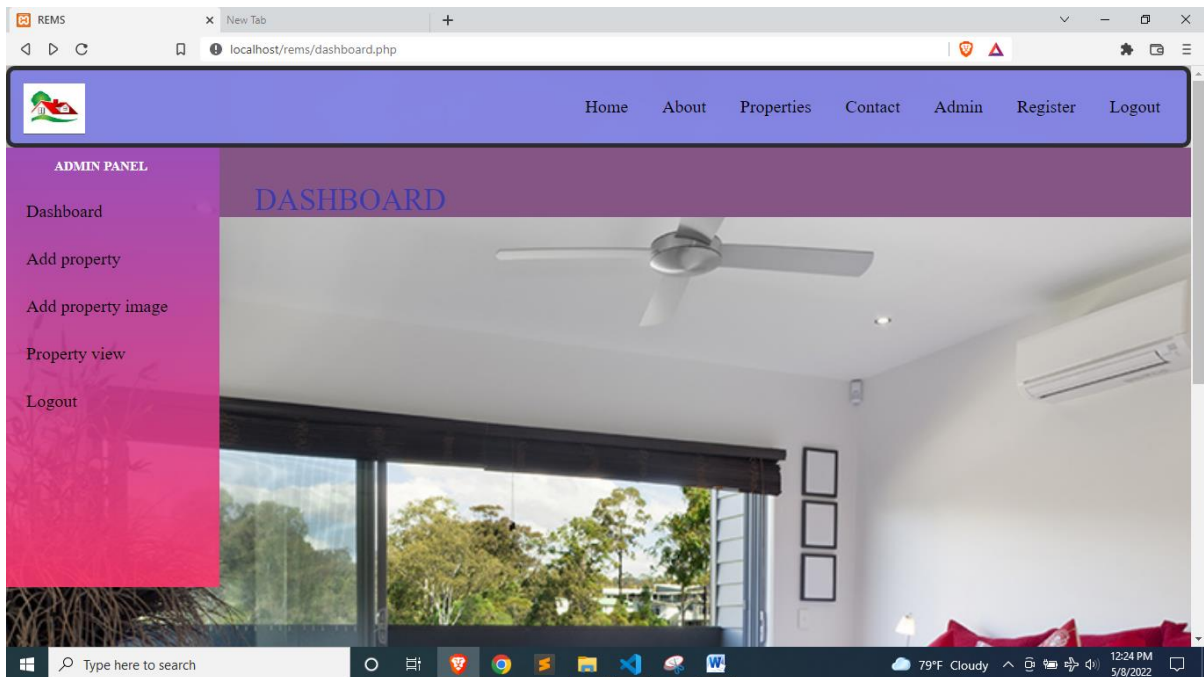


Figure 3-15 Admin Panel Interface

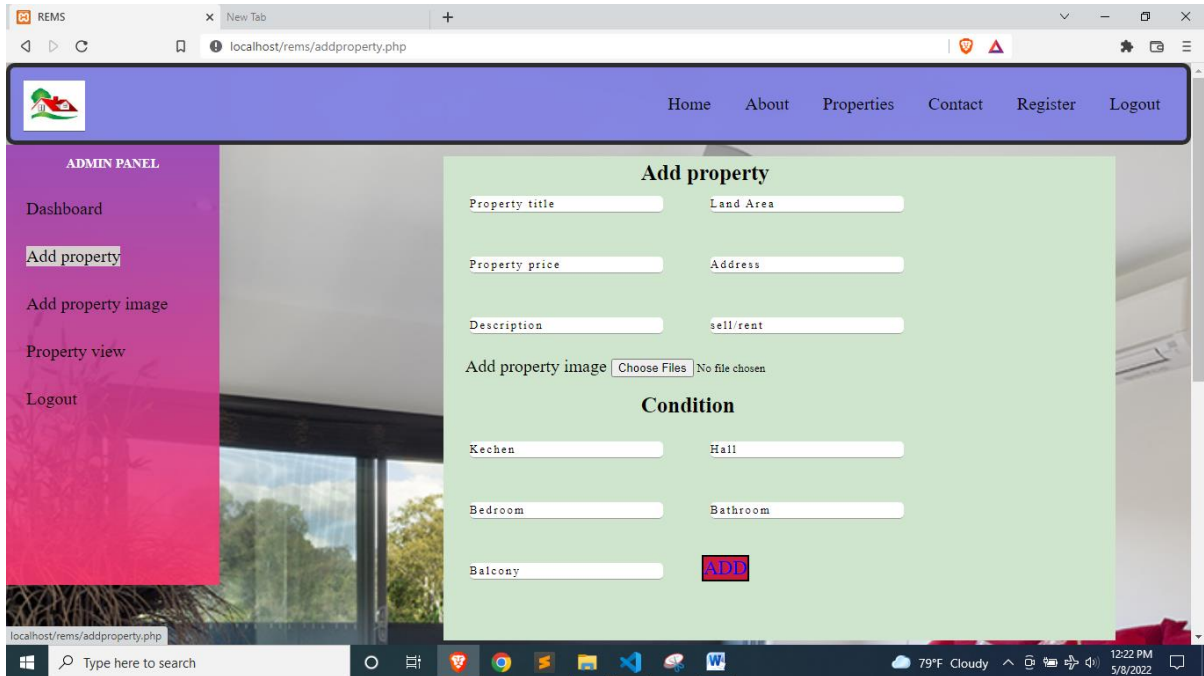


Figure 3-16 Add Properties Interface

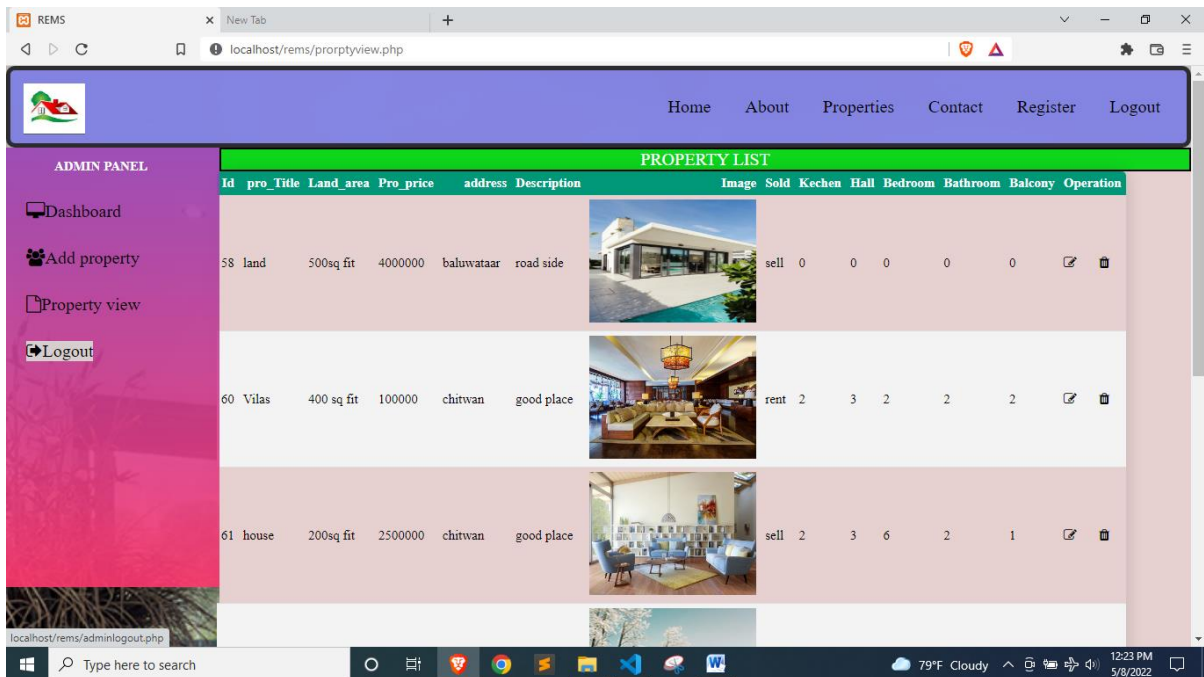


Figure 3-17 Properties View Interface

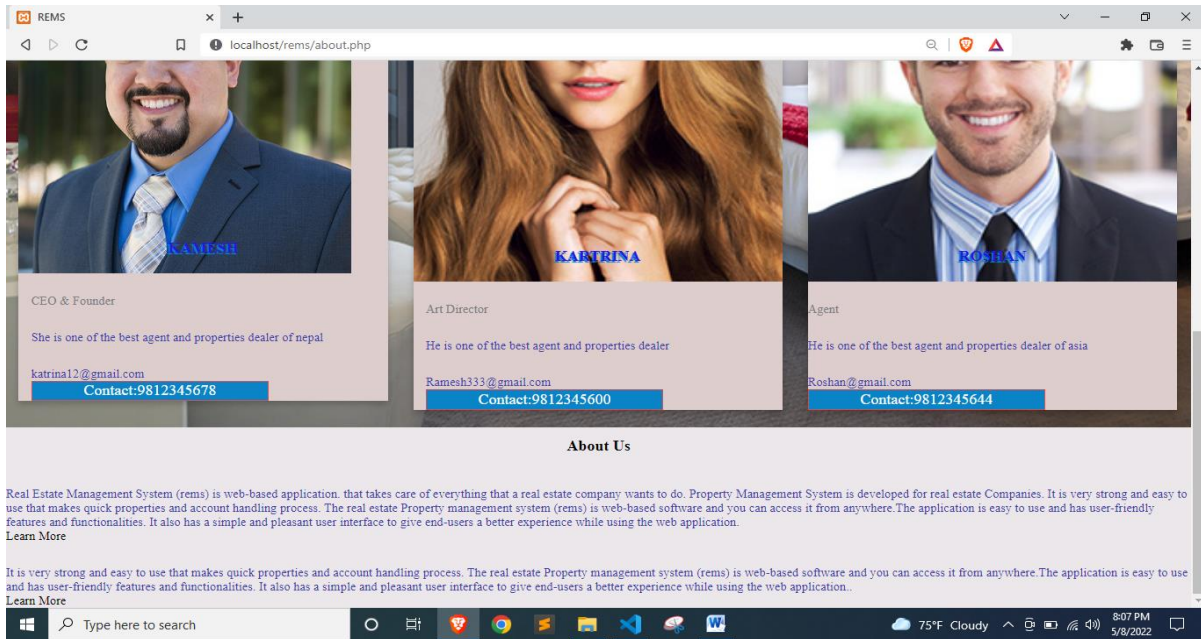


Figure 3-18 About Us Interface

3.2.4 Physical DFD

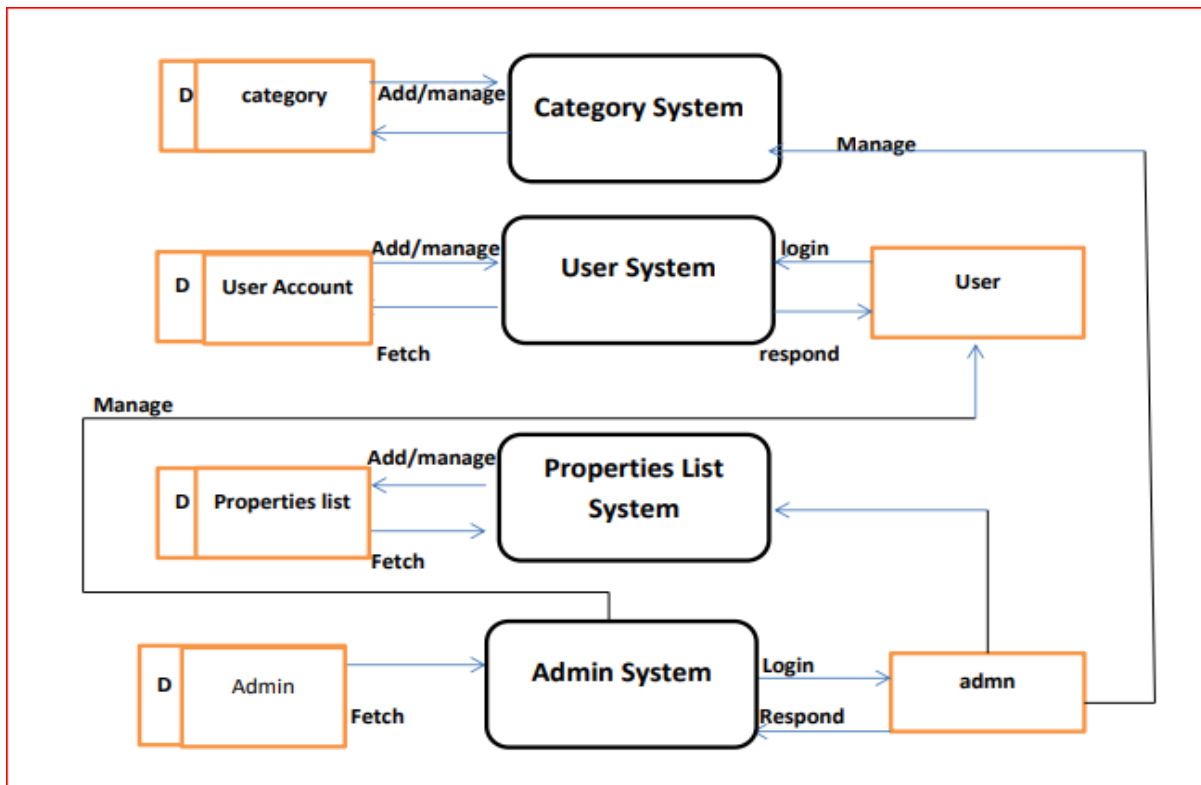


Figure 3-19 Physicl DFD

3.3 System Design

System design represents the overall concept on how the system works. (4) In our system there is only one module i.e. admin module.

Admin Module:

Admin dashboard.

Admin can login and logout.

Admin can add, list, view, update and delete cab category.

Admin can add, view update properties details.

Admin can list, view, update and delete properties details.

Admin can list and view user details.

Admin can manage system information.

Admin can manage account details.

User Module:

User can login and logout.

User can list and view booking details.

User can update own status.

User can manage account details.

User can read 'Welcome' and 'About Us' content.

4. Implementation and Testing

4.1 Implementation

4.1.1 Tools Used (CASE tools, Programming Language, Database Platform)

The Software development life cycle has been used as input in the implementation process. Before coding, I create DFD and ER diagram using draw.io and visual paradigm. The actual implementation has been done using PHP. PHP has been used to interact with the backend database. In this implementation, MySQL Server has been used as the backend. PHP processes the inputs or commands given by the user and translates them in the commands understandable to the backend database. The output produced by the backend database is also handled by PHP which then displayed on the Browser screen.

4.1.2 Implementation Details of Modules (Description of procedures/ functions)

The modules designed for software are as follows:

4.1.2.1 Admin Module

This software can be operated by the admin. So, all the features can be used by the admin.

Features of Admin Module

- Admin can login and logout.
- Admin can add, list, view, update and delete properties.
- Admin can add, view update features of properties.
- Admin can list, view, update and delete user details.
- Admin can list and view sells details.
- Admin can manage system information.
- Admin can manage account details.

4.1.2.2 User Module

The main purpose for developing model is to making easier of buying and selling properties.

Features of User Module

- User can login and logout.
- User can list and view properties details.
- User can manage account details.
- Client can read 'Welcome' and 'About Us' content.

4.2 Testing

Testing is an integral part of the software development process. It is performed at each stage of the SDLC. It ensures that the developed parts conform to the user requirements. It helps to find out whether an input given to the system is well processed or not and output meets the specified objective of the system. It mainly ensures that the system performs as planned. The testing of system was carried out step by step. Testing is performed at key points that are crucial for the working of a system. In hardware system testing is done for individual components that are used to make up a system. In other word testing is a process of Validation and Verification. Validation is the process of checking if the system will meet the customer's actual needs, whereas Verification is concerned whether the system is well-engineered and error free.

During the development process the project encountered many errors which were debugged one by one by identifying problems occurred in each step. The main objectives of system testing are:

- To ensure that during operation the system will perform as per specifications.
- To make sure that the system meets user's requirements during operation.
- To verify that the controls incorporated in the system function as intended.
- To see that when correct inputs are fed to the system, outputs are correct.

4.2.1 System Testing

Integration of different modules comprises a system. Once all the required modules were developed and integrated, the system as a whole was tested to ensure the system was working as per planned. This test is very helpful to determine the problems and bug that have been left out during the integration testing. The test was performed to ensure that the system meet the user requirement. (6)

4.2.2 Integration Testing

In this type of testing, I test various integration of the project module by providing the input. The primary objective is to test the module interfaces in order to ensure that no errors are occurring when one module invokes the other module.

4.2.3 Unit Testing

Unit testing is undertaken when a module has been created and successfully reviewed. In order to test a single module, I need to provide a complete environment i.e., besides the module would require

- The procedures belonging to other modules that the module under test calls
 - Non local data structures that module accesses
 - A procedure to call the functions of the module under test with appropriate parameters
- Unit testing was done on each and every module that is described.

Test for Admin Module

- **Testing admin login form-** This form is used for login of administrator of the system. In this we can enter the username and password if both are correct administration page will open otherwise if any of data is wrong it will get redirected back to the login page and again ask for username and Password.
- **Testing for User login-** In this section the user login to the system. In this if the user enters correct code and password then only they can enter to the system.

4.2.4 Test case for Unit Testing

This involved testing the system in order to correct errors or remove defects that rose. This stage involved testing the source code to make sure that it produced the expected and desired results when subjected to a set of predefined conditions. It was subdivided into three major phases, that is, unit testing, system testing and user acceptance testing.

Table 0-1 Admin Login Test

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TC1	Check Admin Login with valid data	1.Go to site http://localhost/remis/adminlogin.php 2.Enter username 3.Enter password 4.Click Submit	Username=admin Password=admin123	Admin should be login into application	As Expected,	Pass
TC2	Check admin Login with invalid Data	1.Go to site http://localhost/remis/adminlogin.php 2.Enter username 3.Enter password 4.Click Submit	Username=admin Password=123admin	Admin should not be login into application	As Expected,	Fail

Table 0-2 User Login Test

Test Case ID	Test Scenario	Test Steps	Test Data	Expected Results	Actual Results	Pass/Fail
TC3	Check User Login with valid Data	1.Go to site http://localhost/rems/login.php 2.Enter code 3.Enter password 4.Click Submit	Code=202204-00001 Password=123123	Driver should be login into application	As Expected ,	Pass
TC4	Check Driver Login with invalid Data	1.Go to site http://localhost/rems/user/login.php 2.Enter invalid code 3.Enter password 4.Click Submit	Code= 25874562 Password=hrtujrty	Driver should not be login into application	As Expected ,	Fail

5 Conclusion and Future Recommendations

5.1 Lesson Learn/ Outcome

The application can be used for searching property. It is easy to use, since it uses the GUI provided. User friendly screens are provided. The application is easy to use for user to search property and buy their desired property and also satisfied customer without searching visiting many places. It has been thoroughly tested and implemented.

Moreover, the website makes me firmly believe that I can develop other application using simple HTML, CSS and PHP with MySQL database. The requirement for this kind of implementation is minimum and the setup is easy.

5.2 Conclusion

Through the investigation and collecting data I found a lot of problems for searching the properties which is tried to be solved through this software. Hence we have concluded that this project can be beneficial in the daily life and save lot of time. As there is the function of buying properties, searching and accepting the many deals through the mobile phone which make the process more efficient, easy and less time consuming. The maintenance of the record is made efficient, as all the record is stored in access database through which it can be retrieved easily. It allows users to search property online, manage their property and cancel at any point of time. The users will get notified about the agents and his mobile no. so as to communicate with him. Regular updates are provided to the customer so that they are aware of their property, owner details, and property status. I expect this system to be deployed gradually, starting from urban areas to the city area and all over the country.

5.3 Future Recommendations

Software building is never ending process and continues the life of the software as per the changing needs of the user. The project is no doubt has been developed keeping in mind easy modification and enhancement that may be required from time to time. However, there are many scopes to be modified in this software. The additional scopes that can be modified are:

In future, the system will be developed and functionalities will be added as per the user requirements.

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